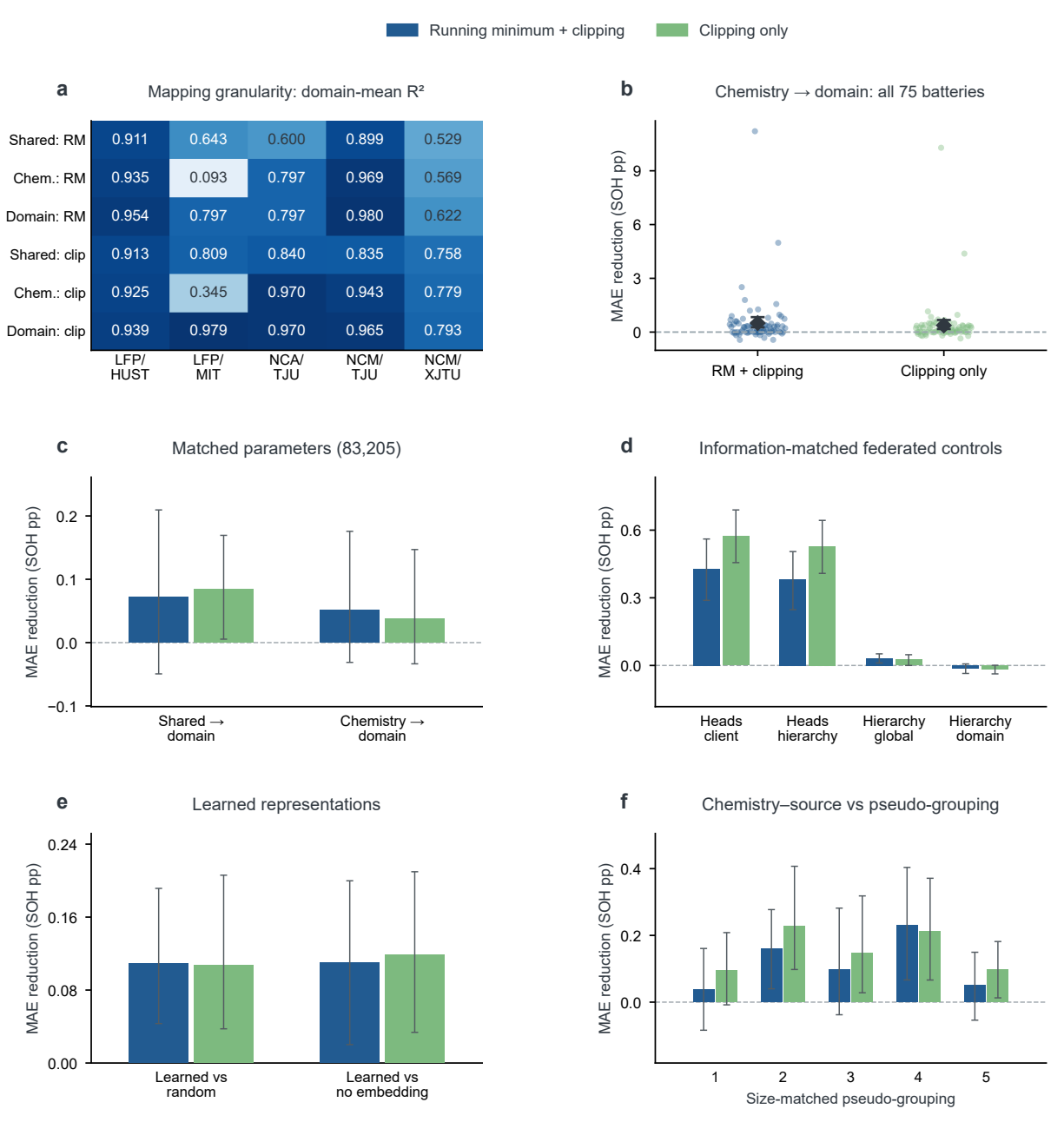

Running minimum + clipping
Clipping only
a
Mapping granularity: domain-mean R²
Shared: RM
Chem.: RM
Domain: RM
Shared: clip
Chem.: clip
Domain: clip
0.911
0.643
0.600
0.899
0.529
0.935
0.093
0.797
0.969
0.569
0.954
0.797
0.797
0.980
0.622
0.913
0.809
0.840
0.835
0.758
0.925
0.345
0.970
0.943
0.779
0.939
0.979
0.970
0.965
0.793
LFP/ HUST
LFP/ MIT
NCA/ TJU
NCM/ TJU
NCM/ XJTU
b
Chemistry → domain: all 75 batteries
MAE reduction (SOH pp)
9
6
3
0
RM + clipping
Clipping only
c
Matched parameters (83,205)
MAE reduction (SOH pp)
0.2
0.1
0.0
−0.1
Shared → domain
Chemistry → domain
d
Information-matched federated controls
MAE reduction (SOH pp)
0.6
0.3
0.0
Heads client
Heads hierarchy
Hierarchy global
Hierarchy domain
e
Learned representations
MAE reduction (SOH pp)
0.24
0.16
0.08
0.00
Learned vs random
Learned vs no embedding
f
Chemistry–source vs pseudo-grouping
MAE reduction (SOH pp)
0.4
0.2
0.0
1
2
3
4
5
Size-matched pseudo-grouping